NNWA Nutrition and Wellness Academy
पोषण और कल्याण अकादमी

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Protein, *decoded.*

How much you need, where to find it in everyday Indian food, and how to spread it across your day.

A plain-language guide

From NNWA (Nutrition & Wellness Academy). General information for healthy adults, not medical advice.

01

Seven short chapters.

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02 How much you need 05

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How to use this guide

Read it once, then use the workbook in chapter 07 for a week. Every fact comes from a government source, listed in the bibliography on page 23.

Before you start

General information for healthy adults, not medical advice. If you are pregnant, breastfeeding, under 18, older, unwell or training heavily, speak to a qualified healthcare professional.

01

What protein *does.*

Before the numbers, here is what protein does in the body and why food has to supply part of it every day.

01 What protein does
02 How much you need
03 Where to find it
04 Spread it across your day
05 Myths, checked
06 Supplements, carefully
07 Your workbook

01 | Six jobs, every day.

Protein is made of building blocks called amino acids. The body uses about 20 of them. It can make some itself; the rest, called essential (indispensable) amino acids, must come from food.

9

essential amino acids adults must get from food

01 **Builds and repairs**
Protein is needed for growth and for the constant repair and rebuilding of body tissues.

02 **Runs reactions**
Enzymes, which drive digestion and metabolism, are proteins.

03 **Sends signals**
Some hormones, such as insulin, are proteins.

04 **Defends**
Antibodies that help fight infection are proteins.

05 **Carries**
Haemoglobin, the protein in red blood cells, carries oxygen.

06 **Gives energy**
About 4 kcal per gram. Carbohydrate and fat are the main fuels in a balanced diet.

Why daily matters

Body proteins are broken down and rebuilt every day, so a steady supply from meals keeps that cycle running. Unlike fat, the body keeps no dedicated store of protein to draw on.

02

How much do you *need?*

One number covers most healthy adults. This chapter gives you that number, a table to find yours, and the situations where it changes.

01 What protein does

02 How much you need

03 Where to find it

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02 | One number to remember.

0.83 g per kg of body weight, a day

This is the Recommended Dietary Allowance (RDA) for healthy Indian adults set by ICMR-NIN. An RDA covers nearly all healthy people in a group, so many people need slightly less.

QUICK MATHS				
	Weight in kg	**× 0.83** the RDA	**= grams** per day	**60 → 50** kg to g, example

When the number changes

Pregnancy and breastfeeding	Needs go up. ICMR-NIN sets separate, higher amounts for each stage.
Children and teenagers	Needs are set by age and growth stage, not by the adult figure.
Older adults	Protein-rich foods help maintain muscle with age; the right amount is best set individually.
Kidney or liver disease	Some conditions need less protein, not more. Follow your doctor's advice.
Heavy training	Athletes may need more, within limits. See chapter 05.

General information only, not medical advice.

02 | Your weight, your grams.

37 g	**at 45 kg**	a day
42 g	**at 50 kg**	a day
46 g	**at 55 kg**	a day
50 g	**at 60 kg**	a day
58 g	**at 70 kg**	a day
66 g	**at 80 kg**	a day

Daily protein at the 0.83 g per kg RDA, rounded to the nearest gram.

03

Where to *find* it.

Protein is already in Indian kitchens: in dal, curd, paneer, eggs, soya, nuts, fish and chicken. Here is roughly how much each serving brings, and how to pair foods well.

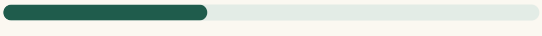
01 What protein does
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03 | Roughly how much, per serving.

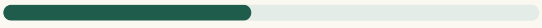
Plant foods

Soya chunks 25 g dry		**13 g**
Cooked dal 1 katori, from 30 g raw		**7 g**
Peanuts 30 g		**7 g**
Roasted chana 30 g		**6 g**
Cooked rice 1 cup, from 50 g raw		**3.5 g**
Roti 1 medium, 30 g atta		**3 g**

Dairy

Paneer 50 g		**9 g**
Milk 200 ml		**6 g**
Curd 1 katori, 150 g		**5 g**

Eggs, fish and meat

Chicken 100 g raw		**21 g**
Fish 100 g raw		**19 g**
Egg 1 large		**6 g**

Approximate, rounded values calculated by NNWA from per-100 g figures in ICMR-NIN's Indian Food Composition Tables. Actual amounts vary with variety, recipe and portion. Bars scale to 13 g; chicken and fish run past the scale.

03 | Better together.

Cereals such as rice and wheat are low in the essential amino acid lysine. Pulses are low in methionine. Eaten in the same meal, each fills the other's gap, which is why so many Indian staples already pair them.

01 **Dal and chawal**
Rice with any dal.

02 **Rajma or chole with rice**
Kidney beans or chickpeas.

03 **Idli or dosa with sambar**
Rice and urad batter, toor dal sambar.

04 **Khichdi**
Rice and moong dal in one pot.

05 **Roti with chana or dal**
Wheat with pulses.

06 **Poha with peanuts**
Flattened rice with a legume.

Complete on their own

Milk, curd, paneer, eggs, fish and meat provide good-quality protein with all nine essential amino acids. Adding a little of any of them to a cereal meal also lifts its protein quality. Soya is often counted here too.

03 | Dal and chawal



PROTEIN INFORMATION	≈ 10 g	1 katori	1 cup	Paired
	protein	cooked dal	cooked rice	pulse + cereal

On the plate

- Dal, from 30 g raw: ≈ 7 g
- Rice, from 50 g raw: ≈ 3.5 g
- Add a katori of curd: ≈ 5 g

Good to know

Rice is low in lysine and dal is low in methionine. Eaten together, each fills the other's gap, so the meal's protein is better used than either food alone.

Try: rajma-chawal, chole with rice, or khichdi.

Approximate values calculated by NNWA from IFCT 2017 per-100 g figures; actual amounts vary by dal, recipe and portion. Photo via Pexels.

03 | Paneer, portioned



PROTEIN INFORMATION	≈ 9 g	50 g	Dairy	Quality
	protein	paneer	food group	good-quality protein

On the plate

Paneer, 50 g: ≈ 9 g

With 2 rotis: + ≈ 6 g

Shown here in a gravy

Good to know

Paneer is a good-quality dairy protein and an easy vegetarian option. It is also rich in fat, so a measured 50 g portion keeps a meal balanced. Grilled, bhurji or in palak works better than heavy, creamy gravies every day.

Try: paneer bhurji with roti, paneer tikka, or palak paneer.

Approximate values calculated by NNWA from IFCT 2017 per-100 g figures; actual amounts vary by milk and recipe. Photo via Pexels.

03 | Eggs, two ways



PROTEIN INFORMATION	≈ 6 g	1 egg	Eggs	Quality
	per large egg	about 50 g	food group	good-quality protein

On the plate

1 large egg: ≈ 6 g

2 eggs at breakfast: ≈ 12 g

With 1 roti: + ≈ 3 g

Good to know

An egg is one of the simplest sources of good-quality protein. The white holds a little over half the protein; the yolk carries most of the fat. Eating the whole egg gives you both.

Try: a boiled egg, egg bhurji with vegetables, or an omelette with roti.

Approximate values calculated by NNWA from IFCT 2017 per-100 g figures. Photo via Pexels.

04

Spread it *across* the day.

A protein food at every meal is easier than one big serving, and it fits how most Indian households already eat.

01 What protein does
02 How much you need
03 Where to find it
04 **Spread it across your day**
05 Myths, checked
06 Supplements, carefully
07 Your workbook

04 | Little and often.

One illustrative day for a 60 kg adult, whose RDA works out to about 50 g. It is an example to adapt, not a meal plan.

BREAKFAST	**A glass of milk with an egg or a handful of roasted chana** Swap in curd, besan chilla or sprouts.	12 g
LUNCH	**Two rotis, a katori of dal, a katori of curd, sabzi** Cereal and pulse in the same meal.	18 g
SNACK	**A small handful of peanuts or sprouts chaat** Easy to carry; watch the portion.	7 g
DINNER	**Rice with rajma, or 50 g paneer with sabzi** Fish or chicken works here too.	13 g

DAY TOTAL, APPROXIMATELY



50 g

Values are approximate, from the serving table on page 09.

05

Myths, *checked.*

Five things people often repeat about protein, set against what ICMR-NIN's guidance for Indians says.

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05 | What the evidence says.

01

~~More protein always means more muscle~~

Training builds muscle. Protein supports it, up to a point.

Muscle grows with resistance exercise plus enough protein and energy. ICMR-NIN's 2024 guidelines note that intakes above about 1.6 g/kg a day add no further training gains.

02

~~Vegetarians cannot get enough protein~~

A varied vegetarian diet can meet the RDA.

Pulses with cereals, plus dairy, nuts and seeds, can cover the needs of most healthy adults.

03

~~Plant protein does not count~~

It counts. It works best in combination.

Single plant foods may be low in one amino acid. Pairing cereals and pulses fills the gap (page 10).

04

~~You need a shake after every workout~~

For most people, regular meals do the job.

ICMR-NIN advises against protein supplements for building muscle (chapter 06).

05

~~Protein is only for gym-goers~~

Everyone needs it, every day.

Growth and repair depend on protein at every age and activity level; see page 04 for its other roles.

Check a claim yourself

Ask who is saying it, whether a named source such as ICMR-NIN backs it, and whether it promises a quick result. Sound nutrition advice rarely does.

06

Supplements, *carefully.*

Protein powders are everywhere. Here is what India's dietary guidelines say about them, and what to check if you are still considering one.

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06 **Supplements, carefully**
07 Your workbook

06 | Start with food.

What the guidelines say

ICMR-NIN's Dietary Guidelines for Indians (2024) advise against protein supplements for building muscle and recommend meeting needs through food combinations. They warn that large amounts of protein powder taken over long periods may be linked to bone mineral loss and kidney damage.

If you are still considering one

01

Speak to a professional first

Especially if you have any medical condition, are pregnant, or are under 18. This is NNWA's general advice.

02

Check the FSSAI licence number

Packaged foods sold in India must show the FSSAI logo and licence number on the label. Buy from a traceable seller.

03

Read the label, not the claims

Labels must list ingredients and nutrition information, including protein, per 100 g and per serving. Check added sugar too.

04

Be wary of big promises

FSSAI rules require claims to be truthful and backed by evidence, and food may not claim to cure or treat disease. Treat big promises as a warning sign.

General information only, not medical advice.

07

Your *workbook.*

Two pages to fill in. Track a week of protein habits, then keep the checklist where you will see it.

01 What protein does
02 How much you need
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07 | A week of habits.

Tick a box each time the habit happens. The ticks in the first column show one example day.

HABIT	M	T	W	T	F	S	S
Protein at breakfast milk, curd, egg, chana	✓						
Protein at lunch dal, rajma, paneer, fish	✓						
Protein at dinner one serving, not the day's worth	✓						
Cereal + pulse together dal-chawal, khichdi, idli-sambar	✓						
Dairy or soya serving curd, milk, paneer, tofu							
Nuts or seeds as a snack a small handful	✓						
Label check before supplements FSSAI number, serving, sugar	✓						

Easiest meal this week

One change for next week

07 | Six habits to keep.

- [ ] **I know my number** — ONCE
  Worked out below, and checked with a professional if page 06 applies to me.
- [ ] **Protein at every meal** — DAILY
  Breakfast, lunch and dinner each include a protein food.
- [ ] **A cereal and pulse together** — DAILY
  Dal-chawal, khichdi, idli-sambar or roti-chana.
- [ ] **A dairy or soya serving** — DAILY
  Curd, milk, paneer, tofu or soya chunks.
- [ ] **Nuts or seeds as a snack** — MOST DAYS
  A small handful, not the whole packet.
- [ ] **Label check before any supplement** — EVERY TIME
  FSSAI number, serving size, added sugar, professional advice.

My weight _____ **kg × 0.83 =** _____ *grams a day*

Bibliography.

Every fact, figure and recommendation in this guide is drawn from the government publications below. Serving values are NNWA's rounded calculations from source 3.

1 **ICMR-NIN. Nutrient Requirements for Indians: Recommended Dietary Allowances and Estimated Average Requirements. ICMR-National Institute of Nutrition, 2020.**
Used on pages 04, 06, 07, 10, 11, 15, 17.

2 **ICMR-NIN. Dietary Guidelines for Indians. ICMR-National Institute of Nutrition, 2024.**
Used on pages 04, 06, 10, 11, 12, 13, 17, 19.

3 **ICMR-NIN. Indian Food Composition Tables (IFCT). ICMR-National Institute of Nutrition, 2017.**
Used on pages 04, 09, 11, 12, 13, 15.

4 **FSSAI. Food Safety and Standards (Labelling and Display) Regulations, 2020. Food Safety and Standards Authority of India.**
Used on page 19.

5 **FSSAI. Food Safety and Standards (Advertising and Claims) Regulations, 2018. Food Safety and Standards Authority of India.**
Used on page 19.

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Important

This guide is general educational information for healthy adults. It is not medical advice and does not diagnose, treat or prevent any condition. If you are pregnant or breastfeeding, under 18, an older adult, training heavily, or living with kidney, liver or any other medical condition, consult a qualified healthcare professional before changing your diet. NNWA (Nutrition & Wellness Academy) is an education provider.

Understand the science on your *plate.*

NNWA (Nutrition & Wellness Academy) is a 100% online academy for nutrition, wellness and fitness certification. This guide is one of a series of short, sourced nutrition basics.

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